redistributed since Q-value estimation considers only the mean. We used the Markov property

$$\begin{aligned} \mathrm{E}_\pi \left[\tilde{G}_0 \mid s_0, a_0, \ldots, s_T, a_T \right] &= \mathrm{E}_\pi \left[\sum_{t=0}^{T} \tilde{R}_{t+1} \mid s_0, a_0, \ldots, s_T, a_T \right] \\ &= \mathrm{E} \left[\tilde{R}_{T+1} \mid s_0, a_0, \ldots, s_T, a_T \right] \\ &= \mathrm{E} \left[\tilde{R}_{T+1} \mid s_T, a_T \right] . \end{aligned} \tag{A57}$$

Redistributing the expectation reduces the variance of estimators since the variance of the random variable is already factored out.

We assume a delayed reward MDP $\tilde{\mathcal{P}}$ with reward

$$\tilde{R}_t = \begin{cases} 0 \,, & \text{for } t \leqslant T \\ \tilde{R}_{T+1} \,, & \text{for } t = T+1 \,, \end{cases} \tag{A58}$$

where $\tilde{R}_t = 0$ means that the random variable $\tilde{R}_t$ is always zero. The expected reward at the last time step is

$$\tilde{r}(s_T, a_T) = \mathrm{E} \left[\tilde{R}_{T+1} \mid s_T, a_T \right] , \tag{A59}$$

which is also the expected return. Given a state-action sequence $(s_0, a_0, \ldots, s_T, a_T)$, we want to redistribute either the realization $\tilde{r}_{T+1}$ of the random variable $\tilde{R}_{T+1}$ or its expectation $\tilde{r}(s_T, a_T)$,

A2.6.1 Optimal Reward Redistribution

The main goal in this paper is to derive an SDP via reward redistribution that has zero expected future rewards. Consequently the SDP has no delayed rewards. To measure the amount of delayed rewards, we define the expected sum of delayed rewards $\kappa(m, t-1)$.

Definition A10. *For $1 \leqslant t \leqslant T$ and $0 \leqslant m \leqslant T - t$, the expected sum of delayed rewards at time $(t-1)$ in the interval $[t+1, t+m+1]$ is defined as*

$$\kappa(m, t-1) = \mathrm{E}_\pi \left[\sum_{\tau=0}^{m} R_{t+1+\tau} \mid s_{t-1}, a_{t-1} \right] . \tag{A60}$$

The Bellman equation for Q-values becomes

$$q^\pi(s_t, a_t) = r(s_t, a_t) + \kappa(T-t-1, t) \,, \tag{A61}$$

where $\kappa(T-t-1, t)$ is the expected sum of future rewards until sequence end given (s_t, a_t), that is, in the interval $[t+2, T+1]$. We aim to derive an MDP with $\kappa(T-t-1, t) = 0$, which gives $q^\pi(s_t, a_t) = r(s_t, a_t)$. In this case, learning the Q-values reduces to estimating the average immediate reward $r(s_t, a_t) = \mathrm{E}\left[R_{t+1} \mid s_t, a_t\right]$. Hence, the reinforcement learning task reduces to computing the mean, e.g. the arithmetic mean, for each state-action pair (s_t, a_t). Next, we define an optimal reward redistribution.

Definition A11. *A reward redistribution is optimal, if $\kappa(T-t-1, t) = 0$ for $0 \leqslant t \leqslant T-1$.*

Next theorem states that in general an MDP with optimal reward redistribution does not exist, which is the reason why we will consider SDPs in the following.

Theorem A3. *In general, an optimal reward redistribution violates the assumption that the reward distribution is Markov, therefore the Bellman equation does not hold.*

Proof. We assume an MDP $\tilde{\mathcal{P}}$ with $\tilde{r}(s_T, a_T) \neq 0$ and which has policies that lead to different expected returns at time $t = 0$. If all reward is given at time $t = 0$, all policies have the same expected return at time $t = 0$. This violates our assumption, therefore not all reward can be given at $t = 0$. In vector and matrix notation the Bellman equation is

$$\boldsymbol{q}_t^\pi = \boldsymbol{r}_t + \boldsymbol{P}_{t \to t+1} \, \boldsymbol{q}_{t+1}^\pi \,, \tag{A62}$$

where $\boldsymbol{P}_{t \to t+1}$ is the row-stochastic matrix with $p(s_{t+1} \mid s_t, a_t)\pi(a_{t+1} \mid s_{t+1})$ at positions $((s_t, a_t), (s_{t+1}, a_{t+1}))$. An optimal reward redistribution requires the expected future rewards to be zero:

$$\boldsymbol{P}_{t \to t+1} \, \boldsymbol{q}_{t+1}^\pi = \boldsymbol{0} \tag{A63}$$

and, since optimality requires $\boldsymbol{q}^{\pi}_{t+1} = \boldsymbol{r}_{t+1}$, we have

$$\boldsymbol{P}_{t \to t+1} \, \boldsymbol{r}_{t+1} \;=\; \boldsymbol{0} \;, \tag{A64}$$

where $\boldsymbol{r}_{t+1}$ is the vector with components $\tilde{r}(s_{t+1}, a_{t+1})$. Since (i) the MDPs are return-equivalent, (ii) $\tilde{r}(s_T, a_T) \neq 0$, and (iii) not all reward is given at $t = 0$, an $(t+1)$ exists with $\boldsymbol{r}_{t+1} \neq \boldsymbol{0}$. We can construct an MDP $\tilde{\mathcal{P}}$ which has (a) at least as many state-action pairs (s_t, a_t) as pairs (s_{t+1}, a_{t+1}) and (b) the transition matrix $\boldsymbol{P}_{t \to t+1}$ has full rank. $\boldsymbol{P}_{t \to t+1} \boldsymbol{r}_{t+1} = \boldsymbol{0}$ is now a contradiction to $\boldsymbol{r}_{t+1} \neq 0$ and $\boldsymbol{P}_{t \to t+1}$ has full rank. Consequently, simultaneously ensuring Markov properties and ensuring zero future return is in general not possible. □

For a particular π, the next theorem states that an optimal reward redistribution, that is $\kappa = 0$, is equivalent to a redistributed reward which expectation is the difference of consecutive Q-values of the original delayed reward. The theorem states that an optimal reward redistribution exists but we have to assume an SDP $\mathcal{P}$ that has a second order Markov reward redistribution.

Theorem A4. *We assume a delayed reward MDP $\tilde{\mathcal{P}}$, where the accumulated reward is given at sequence end. An new SDP $\mathcal{P}$ is obtained by a second order Markov reward redistribution, which ensures that $\mathcal{P}$ is return-equivalent to $\tilde{\mathcal{P}}$. For a specific π, the following two statements are equivalent:*
(I) $\kappa(T-t-1, t) = 0$*, i.e. the reward redistribution is optimal,*

(II) $$\mathrm{E}\left[R_{t+1} \mid s_{t-1}, a_{t-1}, s_t, a_t\right] \;=\; \tilde{q}^{\pi}(s_t, a_t) \;-\; \tilde{q}^{\pi}(s_{t-1}, a_{t-1}) \;. \tag{A65}$$

Furthermore, an optimal reward redistribution fulfills for $1 \leqslant t \leqslant T$ and $0 \leqslant m \leqslant T - t$:

$$\kappa(m, t-1) \;=\; 0 \;. \tag{A66}$$

Proof. PART (I): we assume that the reward redistribution is optimal, that is,

$$\kappa(T-t-1, t) \;=\; 0 \;. \tag{A67}$$

The redistributed reward R_{t+1} is second order Markov. We abbreviate the expected R_{t+1} by h_t:

$$\mathrm{E}\left[R_{t+1} \mid s_{t-1}, a_{t-1}, s_t, a_t\right] \;=\; h_t \;. \tag{A68}$$

The assumptions of Lemma A3 hold for for the delayed reward MDP $\tilde{\mathcal{P}}$ and the redistributed reward SDP $\mathcal{P}$. Therefore for a given state-action sub-sequence $(s_0, a_0, \ldots, s_t, a_t)$, $0 \leqslant t \leqslant T$:

$$\mathrm{E}_{\pi}\left[\tilde{G}_0 \mid s_0, a_0, \ldots, s_t, a_t\right] \;=\; \mathrm{E}_{\pi}\left[G_0 \mid s_0, a_0, \ldots, s_t, a_t\right] \tag{A69}$$

with $G_0 = \sum_{\tau=0}^{T} R_{\tau+1}$ and $\tilde{G}_0 = \tilde{R}_{T+1}$. The Markov property of the MDP $\tilde{\mathcal{P}}$ ensures that the future reward from $t+1$ on is independent of the past sub-sequence $s_0, a_0, \ldots, s_{t-1}, a_{t-1}$:

$$\mathrm{E}_{\pi}\left[\sum_{\tau=0}^{T-t} \tilde{R}_{t+1+\tau} \mid s_t, a_t\right] \;=\; \mathrm{E}_{\pi}\left[\sum_{\tau=0}^{T-t} \tilde{R}_{t+1+\tau} \mid s_0, a_0, \ldots, s_t, a_t\right] \;. \tag{A70}$$

The second order Markov property of the SDP $\mathcal{P}$ ensures that the future reward from $t+2$ on is independent of the past sub-sequence $s_0, a_0, \ldots, s_{t-1}, a_{t-1}$:

$$\mathrm{E}_{\pi}\left[\sum_{\tau=0}^{T-t-1} R_{t+2+\tau} \mid s_t, a_t\right] \;=\; \mathrm{E}_{\pi}\left[\sum_{\tau=0}^{T-t-1} R_{t+2+\tau} \mid s_0, a_0, \ldots, s_t, a_t\right] \;. \tag{A71}$$

Using these properties we obtain

$$
\begin{aligned}
\tilde{q}^{\pi}(s_t, a_t) \;&=\; \mathrm{E}_{\pi}\left[\sum_{\tau=0}^{T-t} \tilde{R}_{t+1+\tau} \mid s_t, a_t\right] \qquad \text{(A72)}\\
&=\; \mathrm{E}_{\pi}\left[\sum_{\tau=0}^{T-t} \tilde{R}_{t+1+\tau} \mid s_0, a_0, \ldots, s_t, a_t\right]\\
&=\; \mathrm{E}_{\pi}\left[\tilde{R}_{T+1} \mid s_0, a_0, \ldots, s_t, a_t\right]\\
&=\; \mathrm{E}_{\pi}\left[\sum_{\tau=0}^{T} \tilde{R}_{\tau+1} \mid s_0, a_0, \ldots, s_t, a_t\right]\\
&=\; \mathrm{E}_{\pi}\left[\tilde{G}_0 \mid s_0, a_0, \ldots, s_t, a_t\right]\\
&=\; \mathrm{E}_{\pi}\left[G_0 \mid s_0, a_0, \ldots, s_t, a_t\right]\\
&=\; \mathrm{E}_{\pi}\left[\sum_{\tau=0}^{T} R_{\tau+1} \mid s_0, a_0, \ldots, s_t, a_t\right]\\
&=\; \mathrm{E}_{\pi}\left[\sum_{\tau=0}^{T-t-1} R_{t+2+\tau} \mid s_0, a_0, \ldots, s_t, a_t\right] \;+\; \sum_{\tau=0}^{t} h_{\tau}\\
&=\; \mathrm{E}_{\pi}\left[\sum_{\tau=0}^{T-t-1} R_{t+2+\tau} \mid s_t, a_t\right] \;+\; \sum_{\tau=0}^{t} h_{\tau}\\
&=\; \kappa(T-t-1, t) \;+\; \sum_{\tau=0}^{t} h_{\tau}\\
&=\; \sum_{\tau=0}^{t} h_{\tau} \;.
\end{aligned}
$$

We used

$$
\kappa(T-t-1, t) \;=\; \mathrm{E}_{\pi}\left[\sum_{\tau=0}^{T-t-1} R_{t+2+\tau} \mid s_t, a_t\right] \;=\; 0 \;. \qquad \text{(A73)}
$$

It follows that

$$
\begin{aligned}
&\mathrm{E}\left[R_{t+1} \mid s_{t-1}, a_{t-1}, s_t, a_t\right] \;=\; h_t \qquad \text{(A74)}\\
&=\; \tilde{q}^{\pi}(s_t, a_t) \;-\; \tilde{q}^{\pi}(s_{t-1}, a_{t-1}) \;.
\end{aligned}
$$

PART (II): we assume that

$$
\begin{aligned}
&\mathrm{E}\left[R_{t+1} \mid s_{t-1}, a_{t-1}, s_t, a_t\right] \;=\; h_t \qquad \text{(A75)}\\
&=\; \tilde{q}^{\pi}(s_t, a_t) \;-\; \tilde{q}^{\pi}(s_{t-1}, a_{t-1}) \;.
\end{aligned}
$$

The expectations $\mathrm{E}_{\pi}\left[. \mid s_{t-1}, a_{t-1}\right]$ like $\mathrm{E}_{\pi}\left[\tilde{R}_{T+1} \mid s_{t-1}, a_{t-1}\right]$ are expectations over all episodes starting in (s_{t-1}, a_{t-1}) and ending in some (s_T, a_T).
First, we consider $m = 0$ and $1 \leqslant t \leqslant T$, therefore $\kappa(0, t-1) = \mathrm{E}_{\pi}\left[R_{t+1} \mid s_{t-1}, a_{t-1}\right]$. Since $\tilde{r}(s_{t-1}, a_{t-1}) = 0$ for $1 \leqslant t \leqslant T$, we have

$$
\begin{aligned}
\tilde{q}^{\pi}(s_{t-1}, a_{t-1}) \;&=\; \tilde{r}(s_{t-1}, a_{t-1}) \;+\; \sum_{s_t, a_t} p(s_t, a_t \mid s_{t-1}, a_{t-1})\, \tilde{q}^{\pi}(s_t, a_t) \qquad \text{(A76)}\\
&=\; \sum_{s_t, a_t} p(s_t, a_t \mid s_{t-1}, a_{t-1})\, \tilde{q}^{\pi}(s_t, a_t) \;.
\end{aligned}
$$